

$$15 + 15 + 20 + 20 + 26 + 14 = 1150$$

= (Sept, 22)

Polynomial Interpolation:

$$h(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

polynomial of n degree.

$a_0, a_1, \dots, a_n$ are constant coefficients

there are $n+1$ coefficients and $n+1$ terms

Polynomials can be thought of as vector space

- you can add vectors
- multiply ~~vectors~~ scalars

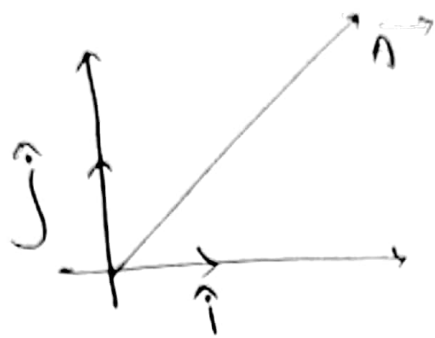
these can be done in polynomials as well.

basis of a vector space:

basis is a set of vectors that spans the space.

Ques.

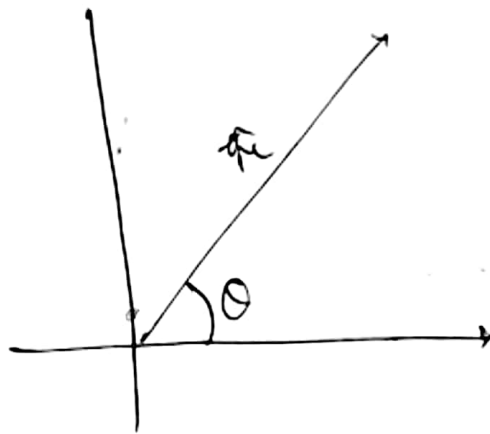
multiple



$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

any vector can be expressed as a combination $\hat{i}, \hat{j}$.

So $\{\hat{i}, \hat{j}\} \rightarrow$ basis.



now, any vector can be expressed by $\{\hat{n}, \hat{\theta}\}$

Interesting thing here is no. of elements in basis corresponds to the dimension of the space.

$$\begin{aligned} \vec{A} &= A_x \hat{i} + A_y \hat{j} \\ &= A_n \hat{n} + A_\theta \hat{\theta} \end{aligned}$$

3 points to remember: ① basis is a set vectors that span the space.

② They are not unique.

③

$f \rightarrow$ Fourier transform of

$$P_2(x) = a_0 \cdot 1 + a_1 x + a_2 x^2$$

$\{1, x, x^2\} \rightarrow$ basis

$\rightarrow$ 3 dimensional space.

$$\therefore P_n(x) = a_0 \cdot 1 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

$\{1, x, x^2, \dots, x^n\} \rightarrow$ natural basis

$(n+1)$ dimensional space.

Function spaces:

they are also vector space. but with infinite dimension.

Fourier Series was

$$f(x) = \sum \left(\tilde{f} \sin(\dots) + f' \cos(\dots) \right)$$

infinite series (infinite no. of sines and cosines)

$\{\sin \dots, \cos \dots\} \rightarrow$ infinite basis

natural basis

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

Polynomial of degree n , belongs a vector space of dimension $n+1$

$$P_n(x) \in V^{n+1}$$

$$\text{function, } f(x) \in V^\infty$$

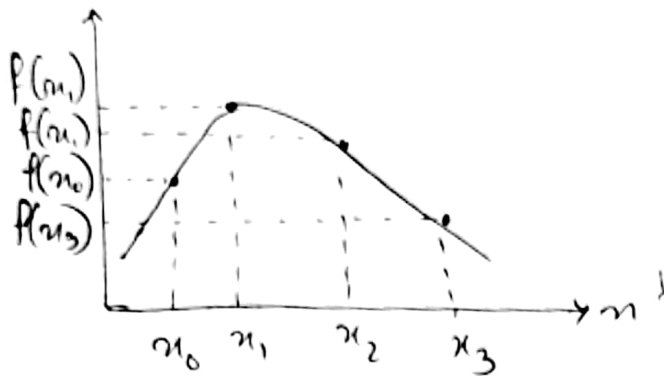
Weierstrass Approximation Theorem:

For a continuous function $f(x)$, on a bounded interval, this is always possible if you take a high enough degree polynomial

for any $f \in C([0,1])$ and any $\epsilon > 0$,
there exists a polynomial such that

$$\max_{0 \leq x \leq 1} |f(x) - p(x)| \leq \epsilon$$

Polynomial Interpolation:

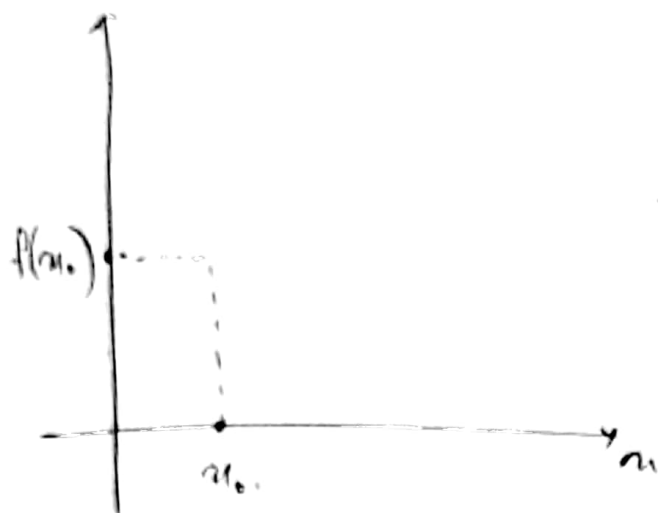


we know, x_0, x_1, x_2, x_3 and corresponding $f(x_0), f(x_1), f(x_2)$ and $f(x_3)$, we don't know the function.

the power of polynomial interpolation,
we can find the polynomial such, that it
goes through these points.

we can say we predict the value of the function
upto some error ϵ .

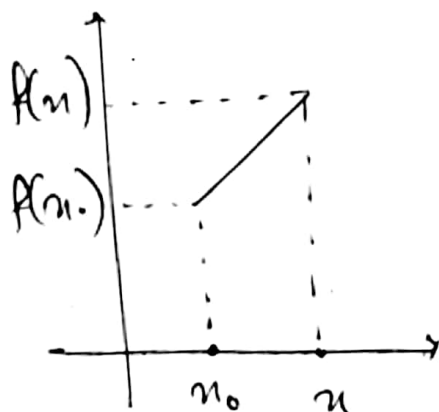
Taylor series:



Let say I know $f(x_0)$, $f'(x_0)$, $f''(x_0)$...
 if I know these, can we predict the value of
 the function at some other point x_n .

Answer is yes.

let say my function is a ~~straie~~ straight line.



$$\frac{f(x) - f(x_0)}{x - x_0} = \tan \theta = f'(x_0)$$

$$\text{or, } f(x) = f'(x_0)(x - x_0) + f(x_0)$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \underbrace{\dots}_{\text{error}}$$

if we don't go to ∞ terms but truncated till any power of x ~~when x~~ , then we get a polynomial

This is polynomial interpolation, with error we did not compute for.

Proof of Taylor series:

A function is a vector in a infinite vector space.

$$\{1, x, x^2, \dots\}$$

$$f(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + a_3(x-x_0)^3 + \dots$$

$$f'(x) = a_1 + 2a_2(x-x_0) + 3a_3(x-x_0)^2 + \dots$$

$$f''(x) = 2a_2 + 3 \times 2 a_3(x-x_0) + \dots$$

$$f'''(x) = 6a_3 + \dots$$

$$\therefore f(x_0) = a_0 \quad [\because \text{all } (x-x_0) = 0]$$

$$\therefore f'(x_0) = a_1 \quad [\because \text{all } (x-x_0) = 0]$$

$$\therefore f''(x_0) = 2a_2 \quad [\because \text{all } (x-x_0) = 0]$$

$$\therefore f'''(x_0) = 3! a_3$$

$$\therefore a_0 = f(x_0)$$

$$a_1 = f'(x_0)$$

$$a_2 = \frac{f''(x_0)}{2!}$$

$$a_3 = \frac{f'''(x_0)}{3!}$$

⋮

$$a_n = \frac{f^n(x_0)}{n!}$$

$$\therefore f(x) = f(x_0) + f'(x_0)(x-x_0)$$

$$+ \frac{f''(x_0)}{2!} (x-x_0)^2 + \frac{f'''(x_0)}{3!} (x-x_0)^3 + \dots$$

Now, $f(x) = \sin(x)$ and $x=0$,

$$\text{then } \sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\text{1st term: } f(0.1) \approx 0.1$$

$$\text{2nd term: } f(0.1) \approx x - \frac{x^3}{3!} \approx 0.1 - \frac{(0.1)^3}{3!} \approx 0.099833$$

$$\text{3rd term: } f(0.1) \approx 0.1 - \frac{(0.1)^3}{3!} + \frac{(0.1)^5}{5!} \approx 0.099833$$

So, more the term, more accurate we get.

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \dots$$

Lets formalize the ~~the~~ error

ξ (greek letter zhi)

Let f be $n+1$ times differentiable on the interval a to b and let $f^{(n)}$ be continuous on $[a, b]$. If $x, x_0 \in [a, b]$ then there exists $\xi \in (a, b)$ such that $f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k \rightarrow$ Taylor polynomial of degree n .
 $+ \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1} \rightarrow$ the Lagrange form of the remainder

$$f(x_0) + \frac{f'(x_0)}{1!} (x-x_0) + \frac{f''(x_0)}{2!} (x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$$

So, error is $\frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1}$

So, error cannot be calculated, but it can be predicted at a bound as $\xi \in (a, b)$

Example: $f(x) = \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$

$$P_6(x)$$

↳ as next term of x^7

$$\therefore f(x) = P_6(x) + \frac{f^{(7)}(\xi)}{7!} (x-x_0)^7$$

$$|f(x) - P_6(x)| = \left| \frac{f^{(7)}(\xi)}{7!} (x-x_0)^7 \right|$$

Now, let $x = 0.1$, $\therefore f^{(7)}(x) = -\sin x$
 $x_0 = 0$

$$|f(0.1) - P_6(0.1)| = \frac{1}{5040} (0.1)^7 |-\sin \xi|$$

$$\sin(\xi) \leq 1$$

$$\therefore |f(0.1) - P_6(0.1)| \leq \frac{1}{5040} (0.1)^7$$

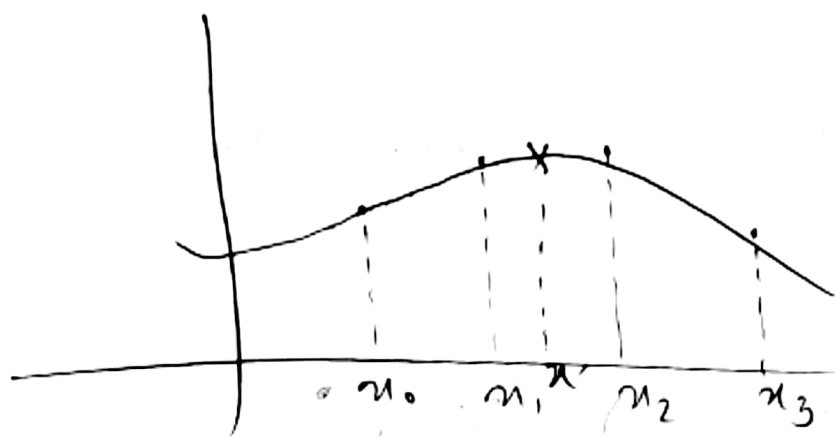
$$|f(0.1) - P_6(0.1)| \leq 1.984 \times 10^{-11}$$

$$1.983 \times 10^{-11}$$

exact

Truncation of an infinite series gives us truncation error.

Polynomial Interpolation:



For interpolation, we will have multiple nodes

(x_0, x_1, x_2, x_3) and we want to find the polynomial

that go through the points.

and find $f(x)$ for any x .

Now,

$$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

$$= \sum_{k=0}^n \cancel{a_k} a_k x^k$$

So, for finding P_n , we need $a_0, a_1, \dots, a_n$

Condition,

$$\text{at the nodes } P_n(x_0) = f(x_0)$$

$$P_n(x_1) = f(x_1)$$

$$P_n(x_2) = f(x_2)$$

$$\cancel{P_n(x_{n-1}) = f(x_{n-1})}$$

$$P_n(x_n) = f(x_n)$$

example:

$$n=1$$

we know $\left. \begin{array}{l} n_0 \rightarrow f(n_0) \\ n_1 \rightarrow f(n_1) \end{array} \right\}$ from this
find a_0, a_1

$$\therefore P_1(n) = a_0 + a_1 n$$

we know,

$$P_1(n_0) = a_0 + a_1 n_0 = f(n_0) \quad \text{--- (i)}$$

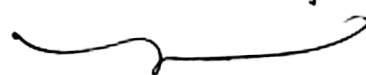
$$P_1(n_1) = a_0 + a_1 n_1 = f(n_1) \quad \text{--- (ii)}$$

we need to get value of 2 values from 2 equations

subtracting we get

$$a_1 (n_0 - n_1) = \underline{f(n_0) - f(n_1)}$$

$$\therefore a_1 = \frac{f(n_0) - f(n_1)}{n_0 - n_1}$$


all known.

putting a_0 in eqⁿ (i) or ii we can get a_1 .

$$a_0 + \frac{f(x_0) - f(x_1)}{x_0 - x_1} x_1 = f(x_1)$$

$$a_0 = \frac{f(x_1) - x_1 \frac{f(x_0) - f(x_1)}{x_0 - x_1}}{x_0 - x_1}$$

$$= \frac{f(x_1)(x_0 - x_1) - x_1 f(x_0) + x_1 f(x_1)}{x_0 - x_1}$$

$$= \frac{x_0 f(x_1) - x_1 f(x_0) - x_1 f(x_1) + x_1 f(x_1)}{x_0 - x_1}$$

$$= \frac{x_0 f(x_1) - x_1 f(x_0)}{x_0 - x_1}$$

$$\therefore p_1(x) = a_0 + a_1 x.$$

But doing this algebra is very hectic.

we can do this using matrix.

$$\begin{pmatrix} 1 & x_0 \\ 1 & x_1 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} f(x_0) \\ f(x_1) \end{pmatrix}$$

$$\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 1 & x_0 \\ 1 & x_1 \end{pmatrix}^{-1} \begin{pmatrix} f(x_0) \\ f(x_1) \end{pmatrix}$$

$$= \frac{1}{\det} \begin{pmatrix} x_1 & -x_0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} f(x_0) \\ f(x_1) \end{pmatrix}$$

HW $\rightarrow$ finish the calculation.

Now for,
 $x_0, x_1, \dots, x_n \rightarrow (n+1)$ nodes

So $(n+1)$ conditions:

So we need $(n+1)$ coeffs

$\therefore$ polynomial is of degree n .

$$P_n(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

conditions:

$$P_n(x_i) = f(x_i)$$

Now, $f(x_0) = a_0 + a_1 x_0 + a_2 x_0^2 + \dots + a_n x_0^n$

$$f(x_1) = a_0 + a_1 x_1 + a_2 x_1^2 + \dots + a_n x_1^n$$

$$f(x_2) = a_0 + a_1 x_2 + a_2 x_2^2 + \dots + a_n x_2^n$$

$$\vdots$$
$$f(x_n) = a_0 + a_1 x_n + a_2 x_n^2 + \dots + a_n x_n^n$$

~~$$a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$~~

~~$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{pmatrix}$$~~

Vandermonde Matrix

A

Need to
find the
co-effs

F

1/

$$VA = F$$

$$A = V^{-1} F$$

catch: V has to be invertible and

inverse exist iff $\det(V) \neq 0$ and $\det(V) = \prod_{0 \leq i < j \leq n} (x_i - x_j)$

$$\det(V) = (x_0 - x_1)(x_1 - x_2) \dots (x_{n-1} - x_n)$$

this is not zero iff $x_i \neq x_j$ if $i \neq j$

i.e. all the nodes has to be distinct.

Let say we have $x_0, x_1, \dots, x_n \rightarrow (n+1)$ nodes

$$\downarrow \quad \downarrow \quad \dots \quad \downarrow$$

$$f(x_0) \quad f(x_1) \quad \dots \quad f(x_n)$$

$\therefore (n+1)$ condition $\Rightarrow n+1$ coeffs

Theorem (Existence / Uniqueness):

Given $(n+1)$ distinct nodes, $x_0, x_1, \dots, x_n$

there is a unique polynomial P_n that interpolates $f(x)$ at the nodes

Proof:

$$p_n \longleftrightarrow q_n$$

$$r_n := p_n - q_n$$

~~r_n should have a degree $\leq n$~~

example:

$$f(x) = \cos(x)$$

$$x_0 = 0, \quad x_1 = \pi/2, \quad x_2 = \pi$$

$$f(x_0) = 1 \quad \text{ess } f(x_1) = 0 \quad f(x_2) = -1$$

3 nodes, so $P_2(x)$

$$P_2(x_0) = a_0 + a_1 x_0 + a_2 x_0^2 = a_0 + 0 + 0$$

$$P_2(x_1) = a_0 + a_1 x_1 + a_2 x_1^2 = a_0 + a_1 \pi/2 + a_2 (\pi/2)^2$$

$$P_2(x_2) = a_0 + a_1 x_2 + a_2 x_2^2 = a_0 + a_1 \pi + a_2 \pi^2$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & \pi/2 & \pi^2/4 \\ 1 & \pi & \pi^2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & \pi/2 & \pi^2/4 \\ 1 & \pi & \pi^2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

CSF 72

graphi

Not rec

image

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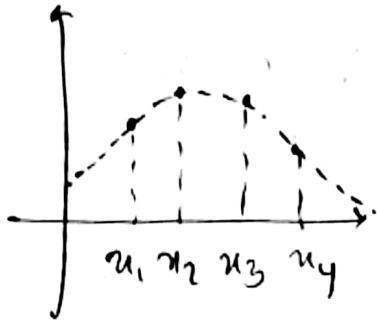
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Lagrange form



We didn't know $f(x)$ but knew some points.

So we found some polynomials such that it goes through the points (-----)

$(n+1)$ nodes $\rightarrow P_n$

Computation complexity is very high for vandermonde matrix.

So we found lagrange form.

basis:

~~We will choose a~~ choose a basis such that

Previously x	$f(x)$
x_0	2
x_1	5
x_2	7
	30
	45
	60

$$\therefore P_2(x) = a_0 x + a_1 x^1 + a_2 x^2$$

$$\text{but now } P_2(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2)$$

l_0, l_1 and l_2 are lagrange basis.

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} \times \frac{x - x_2}{x_0 - x_2}$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} \times \frac{x - x_2}{x_1 - x_2}$$

$$l_2(x) = \frac{x - x_0}{x_2 - x_0} \times \frac{x - x_1}{x_2 - x_1}$$

Example.	Time	Velocity
n_0	15	227.04
n_1	20	362.78
n_2	22.5	517.35

$$P_0(17) = ?$$

$$P_2(n) = l_0(n) f(n_0) + l_1(n) f(n_1) + l_2(n) f(n_2)$$

$$l_0(n) = \frac{n - n_1}{n_0 - n_1} \times \frac{n - n_2}{n_0 - n_2} = \frac{n - 20}{15 - 20} \times \frac{n - 22.5}{15 - 22.5}$$

$$= \frac{2}{75} (n - 20) (n - 22.5)$$

$$l_1(n) = \frac{n - n_0}{n_1 - n_0} \times \frac{n - n_2}{n_1 - n_2} = \frac{n - 15}{20 - 15} \times \frac{n - 22.5}{20 - 22.5}$$

$$= -\frac{2}{25} (n - 15) (n - 22.5)$$

$$I_2(n) = \frac{n - n_0}{n_2 - n_0} \times \frac{n - n_1}{n_2 - n_1} = \frac{n - 15}{22.5 - 15} \times \frac{n - 20}{22.5 - 20}$$

$$= \frac{4}{75} (n - 15)(n - 20)$$

$$\therefore P_2(n) = \frac{2}{75} (n - 20)(n - 22.5) \times 227.04$$

$$- \frac{2}{25} (n - 15)(n - 22.5) \times 362.78$$

$$+ \frac{4}{75} (n - 15)(n - 20) \times 517.35$$

$$\therefore P_2(17) = \frac{2}{75} (17 - 20)(17 - 22.5) \times 227.04$$

$$- \frac{2}{25} (17 - 15)(17 - 22.5) \times 362.78$$

$$+ \frac{4}{75} (17 - 15)(17 - 20) \times 517.35$$

$$f(x) = \cos(x)$$

$$x_0 = -\pi/4$$

$$x_1 = 0$$

$$x_2 = \pi/4$$

$$p_n(x) = ?$$

first! lets think what would be n ?

$\therefore n=2$ as three nodes are given.

$$P_2(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2)$$

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} \times \frac{x - x_2}{x_0 - x_2} = \frac{x - 0}{-\pi/4} \times \frac{x - \pi/4}{-\pi/4 - \pi/4} = \frac{8x}{\pi^2}$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} \times \frac{x - x_2}{x_1 - x_2} = \frac{x + \pi/4}{-\pi/4} \times \frac{x - \pi/4}{0 - \pi/4} = -\frac{16}{\pi^2} (x + \pi/4)$$

$$l_2(x) = \frac{x - x_0}{x_2 - x_0} \times \frac{x - x_1}{x_2 - x_1} = \frac{x + \pi/4}{\pi/4 + \pi/4} \times \frac{x - 0}{\pi/4 - 0} = \frac{8}{\pi^2} (x + \pi/4)$$

$$f(n_0) = 1/\sqrt{2}$$

$$f(n_1) = 1$$

$$f(n_2) = 1/\sqrt{2}$$

$$P_2(n) = \frac{8}{\pi^2} n \left(n - \frac{\pi}{4} \right) \frac{1}{\sqrt{2}} - \frac{16}{\pi^2} \left(n + \frac{\pi}{4} \right) \left(n - \frac{\pi}{4} \right) \\ + \frac{8}{\pi^2} \left(n + \frac{\pi}{4} \right) \frac{1}{\sqrt{2}}$$

$$\left(n - \frac{\pi}{4} \right)$$

$$\left(n + \frac{\pi}{4} \right) \left(n - \frac{\pi}{4} \right)$$

Time	Velocity
0	0
10	227.4
15	362.8
20	517.35
22.5	602.97
30	901.67

~~Q20~~ if mentioned $P_2(16)$, then have to consider only 3 nodes.
 so we will take 3 nodes close to 16.

Newton's divide and difference:

$$P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2) + \dots + a_n(x-x_0)(x-x_1)(x-x_2)\dots(x-x_{n-1})$$

$$a_0 = f[x_0]$$

$$a_1 = f[x_0, x_1]$$

$$a_2 = f[x_0, x_1, x_2]$$

$\vdots$

$$a_n = f[x_0, x_1, x_2, \dots, x_n]$$

$$\begin{aligned} \therefore P_4(x) &= f[x_0] \cancel{(x-x_0)} + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2] \\ &\quad (x-x_0)(x-x_1) \cancel{(x-x_2)} + f[x_0, x_1, x_2, x_3] \\ &\quad (x-x_0)(x-x_1)(x-x_2) + f[x_0, x_1, x_2, x_3, x_4] \\ &\quad (x-x_0)(x-x_1)(x-x_2)(x-x_3) \end{aligned}$$

Example:

	n	$f(n)$
n_0	-1	5
n_1	0	1
n_2	1	3
n_3	2	11
n_4	4	20

$$P_3(n) = f[n_0] + f[n_0, n_1](n - n_0) + f[n_0, n_1, n_2](n - n_0)(n - n_1) + f[n_0, n_1, n_2, n_3](n - n_0)(n - n_1)(n - n_2) \bullet$$

$$n_0 = -1 \quad f[n_0] = 5$$

$$f[n_0, n_1] = \frac{1 - 5}{0 - (-1)} = -4$$

$$n_1 = 0 \quad f[n_1] = 1$$

$$f[n_0, n_1, n_2] = \frac{2 - (-4)}{1 - (-1)} = \frac{2 - (-4)}{1 - (-1)} = \frac{6}{2} = 3$$

$$f[n_1, n_2] = \frac{3 - 1}{1 - 0} = 2$$

$$n_2 = 1 \quad f[n_2] = 3$$

$$f[n_0, n_1, n_2, n_3] = \frac{11 - 3}{2 - 0} = \frac{8}{2} = 4$$

$$f[n_2, n_3] = \frac{11 - 3}{2 - 1} = 8$$

$$n_3 = 2 \quad f[n_3] = 11$$

$$f[n_2, n_3, n_4] = \frac{20 - 11}{4 - 2} = \frac{9}{2}$$

$$f[n_3, n_4] = \frac{20 - 11}{4 - 2} = \frac{9}{2}$$

$$n_4 = 4 \quad f[n_4] = 20$$

$$\therefore P_3(n) = 5 + (-4)(n-n_1) + 3(n-n_1)(n-n_2) + 0(n-n_1)(n-n_2)(n-n_3)$$

Now, if n_4 comes, we need to incorporate to the previous polynomial.

If we did the same for Lagrange l_0, l_1 and l_2 and l_3 would have to be calculated again.

But in Newton's method it is very easy.

$$f[n_1, n_2, n_3] = \frac{3-3}{2+1} = 0$$

$$f[n_1, n_2, n_3, n_4] = \frac{-\frac{7}{6} - 3}{4-0} = \frac{-15}{24}$$

$$f[n_1, n_2, n_3, n_4] = \frac{-\frac{15}{24} - 0}{4+1} = -\frac{5}{24}$$

$$\therefore P_4(n) = 5 + (-4)(n-n_1) + 3(n-n_1)(n-n_2) + 0(n-n_1)(n-n_2)(n-n_3) + -\frac{5}{24}(n-n_1)(n-n_2)(n-n_3)$$

Time	Velocity
10	227.4
15	362.8
20	517.35
30	901.67

a) $P_n(n)$ using Newton's method

b) $P_n(17) = ?$

c) $P_n(n)$ for $n_3 = 30$, $f(n_3) = 1.67$

$$\therefore P_2(n) = 227.4 + 27.08$$

b) Putting $n = 17$

c)

$$a) P_2(n) = f[n_0] + f[n_0, n_1](n - n_0) + f[n_0, n_1, n_2](n - n_0)(n - n_1) \\ + \cancel{f[n_0, n_1, n_2, n_3](n - n_0)(n - n_1)(n - n_2)}$$

$$n_0 = 10 \quad f[n_0] = 227.4$$

$$f[n_0, n_1] = \frac{362.8 - 227.4}{15 - 10} = 27.08$$

$$n_1 = 15 \quad f[n_1] = 362.8$$

$$f[n_0, n_1, n_2] = \frac{517.35 - 362.8}{20 - 15} = 13.91$$

$$n_2 = 20 \quad f[n_2] = 517.35$$

$$f[n_0, n_1, n_2, n_3] = \frac{901.67 - 517.35}{30 - 20} = 38.432$$

$$n_3 = 30 \quad f[n_3] = 901.67$$

$$f[n_0, n_1, n_2] = \frac{13.91 - 27.08}{20 - 10} = -0.383$$

$$f[n_0, n_1, n_2, n_3]$$

d

$$\therefore P_2(n) = 227.4 + 27.08(n - x_0) + 0.383(n - x_0)(n - x_1).$$

1.67

b) Putting $n = 17$, $\rightarrow$

c)

= 0.383.

$$f[x_0, x_1, x_2, x_3]:$$

Cauchy's theorem

$$\left| f(n) - P_n(n) \right| = \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} (n-n_0)(n-n_1)\dots(n-n_n) \right|$$

Example:

Given, $f(x) = \cos(x)$

n values $\rightarrow$

$\left\{ -\frac{\pi}{4}, 0, \frac{\pi}{4} \right\}$

$\uparrow \quad \uparrow \quad \uparrow$

$x_0 \quad x_1 \quad x_2$

Now, calculate the upper bound of error using Cauchy's theorem.

[Interval of $\mathbb{R}$ may or may not be given]

$$|f(z) - p_2(z)| = \frac{f^3(z)}{3!} (z + \pi/4)(z - 0)(z - \pi/4)$$

$$= \left| \frac{\sin(\frac{\pi}{4})}{3!} \right| \left| (n + \pi/4)(n-0)(n-\pi/4) \right|$$

$$\therefore \left| \frac{\sin(1)}{3!} \right| \left| \binom{n+\pi/4}{n-0} \right| \left| \binom{n-\pi/4}{n} \right|$$

NB: for any function, if get 0 after differentiating, then point give the maximum value

$$\omega(n) = \left(n - \frac{\pi}{4}\right) - \left(n - \frac{\pi}{4}\right) \\ = n - \frac{\pi^2}{16}$$

$$= n^3 - n \frac{\pi^2}{16} \quad \text{--- (i)}$$

$$\omega'(n) = 3n^2 - \frac{\pi^2}{16} = 0.$$

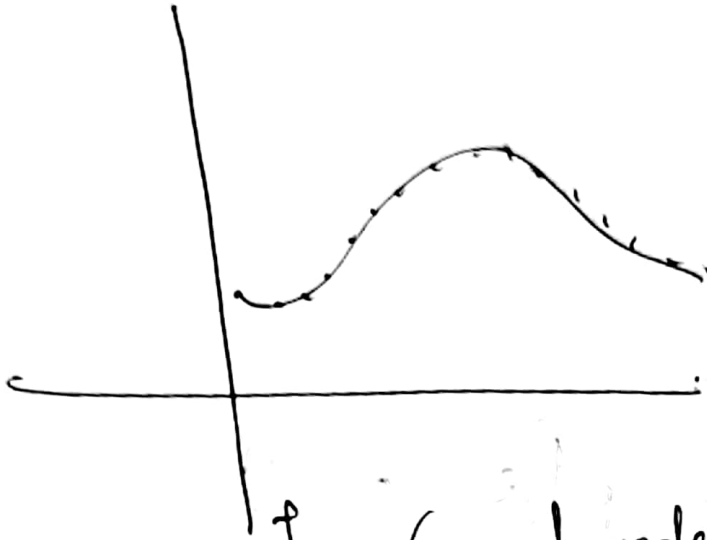
$$\therefore n^2 = \sqrt{\frac{12}{3 \times 16}}$$

$$= 1 \frac{1}{4} (3)$$

Now, putting $u = \frac{+A}{A/B}$ in (i)

x	$ \omega(x) $
$-\pi/4\sqrt{3}$	0.186
$+\pi/4\sqrt{3}$	0.186
1	0.383

Chebyshev Nodes



if n (no. of nodes) $\rightarrow \infty$,

then error should become 0.

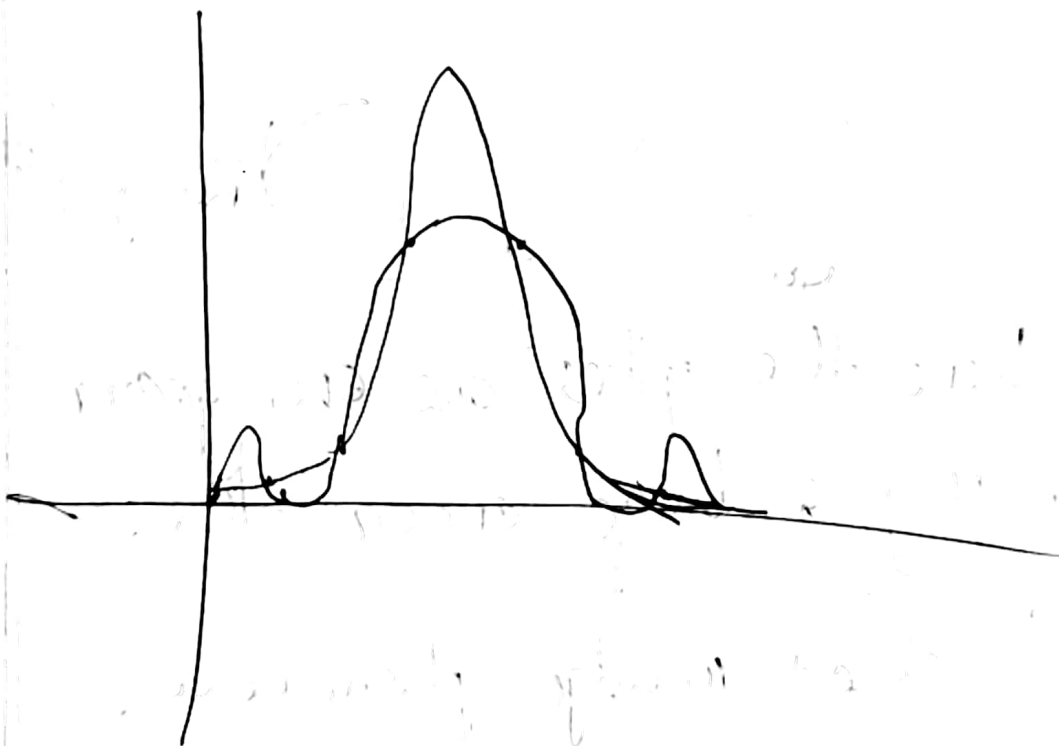
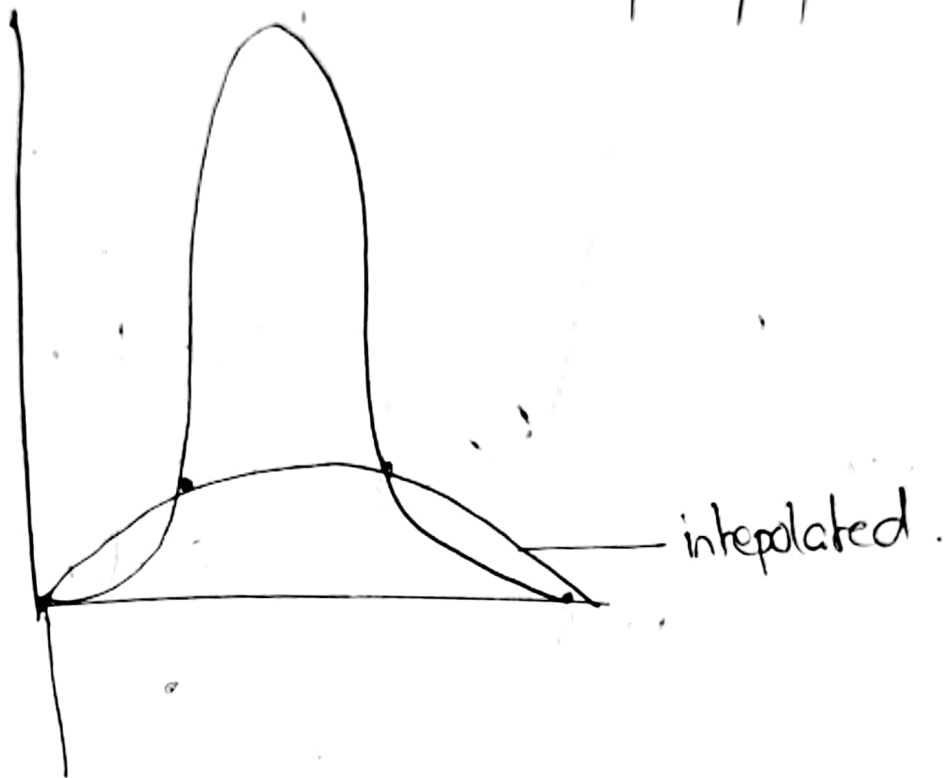
but for some function this is not true.

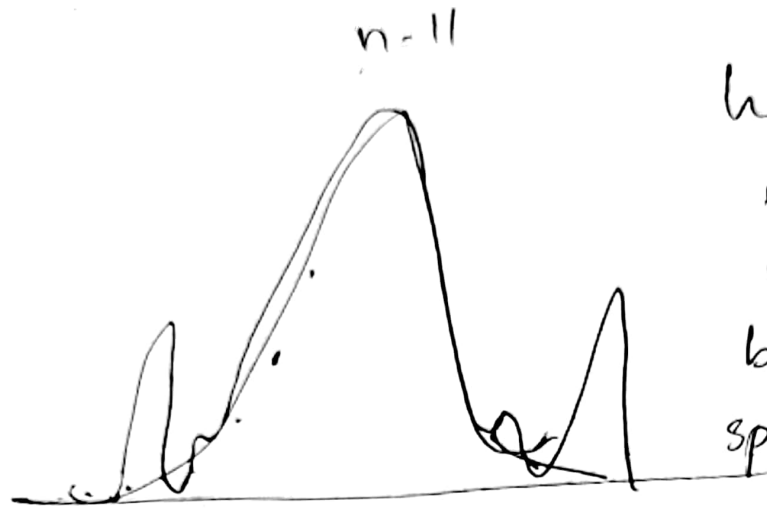
Runge (1901)

$$f(x) = \frac{1}{1 + 25x^2} \text{ on } [-1, 1]$$

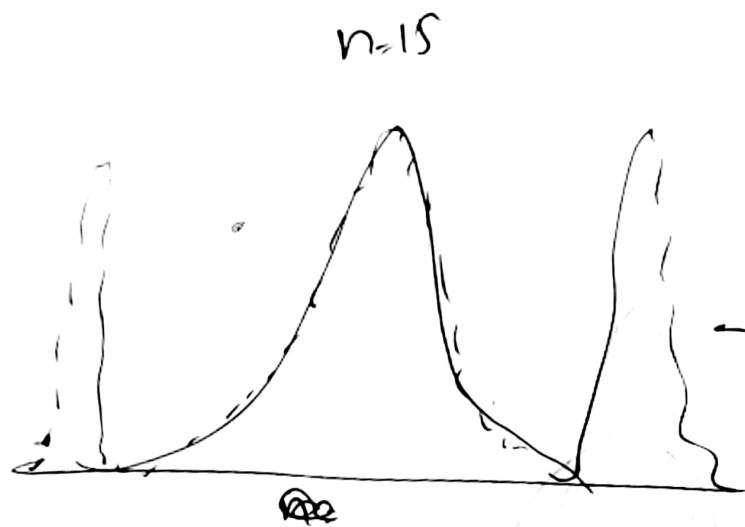
$n=3$

equally spaced nodes.





here the
error is
very close
but but
spikes at the
end.



Runge phen

here the spikes are even greater
creating a huge error. This
is called Runge phenomena

For prod look at paper by Runge.

beyond scope of course.

If $n \rightarrow \infty$, error $\rightarrow \infty$

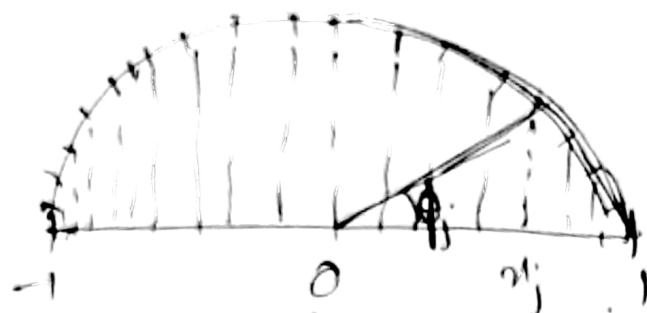
$$\omega(n) = (n-x_0)(n-x_1) \dots (n-x_n)$$

→ we saw this in Cauchy's theorem, that this was part of the error.

now, for $\omega_n \approx n!$ (for equally spaced nodes)

So, $\omega(n)$ diverges when there is equally spaced node.

So, one way to counter this is taking unequal spaced nodes. Now as errors are at the ~~only~~ corners we take more nodes at the corners



so we take a semi circle on the number line with equally angled division. project them on the number line.

$$\phi_j = \frac{(2j+1)\pi}{2(n+1)} \quad j = 0 \dots n.$$

$$x_j = \text{radius} \times \cos(\phi_j) \\ = \cos \left[\frac{(2j+1)\pi}{2(n+1)} \right] \quad j = 0 \dots n$$

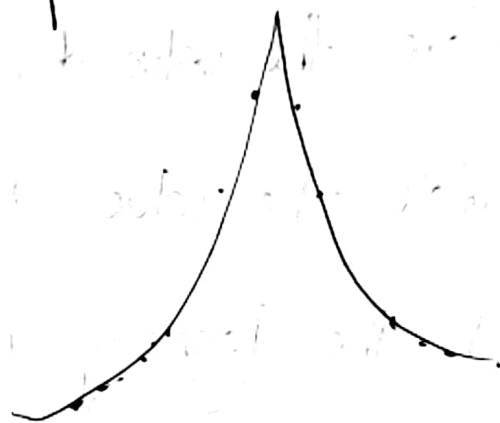
$$f(n) = \frac{1}{1 + 25n^2}, \quad n = 3$$

$$n_1 = \cos \frac{(2 \times 0 + 1)\pi}{2 \times 4} \Rightarrow n_1 = \cos \pi/8$$

not equal $\leftarrow n_1 = \cos \frac{3\pi}{8}$ $\leftarrow$ equal

$$n_2 = \cos \frac{5\pi}{8}$$

$$n_3 = \cos \frac{7\pi}{8}$$



Given Runge function,

$$f(x) = \frac{3}{4 + 9x^2} \quad [-4, 4]$$

$$n = 3$$

- Calculate the ~~value~~ of equal angled points of ϕ
- Calculate the value of Chebyshev nodes (n value)
- Find the Lagrange basis, $l_2(x)$

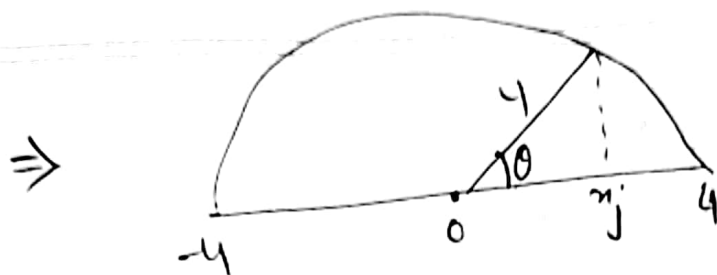
$$\Rightarrow \phi_j = \frac{(2j+1)\pi}{2(n+1)} \quad j = 0, \dots, 3$$

$$\phi_0 = \frac{\pi}{8}$$

$$\phi_1 = \frac{3\pi}{8}$$

$$\phi_2 = \frac{5\pi}{8}$$

$$\phi_3 = \frac{7\pi}{8}$$



$$\cos \theta_j = \frac{x_j}{4}$$

$$x_j = 4 \cos \theta_j$$

$$x_0 = 4 \times \cos \frac{\pi}{8}$$

$$x_1 = 4 \times \cos \frac{3\pi}{8}$$

$$x_2 = 4 \times \cos \frac{5\pi}{8}$$

$$x_3 = 4 \times \cos \frac{7\pi}{8}$$

$$\Rightarrow l_2(x) = \frac{x - x_0}{x_2 - x_0} \times \frac{x - x_1}{x_2 - x_1}$$

$$= \frac{x - 4 \cos \frac{\pi}{8}}{4 \cos \frac{5\pi}{8} - 4 \cos \frac{\pi}{8}} \times \frac{x - 4 \cos \frac{3\pi}{8}}{4 \cos \frac{5\pi}{8} - 4 \cos \frac{3\pi}{8}}$$

x	$f(x)$

$$x \quad \frac{n - 4 \cos \frac{7\pi}{8}}{\quad}$$

$$\text{Arccos } \frac{5\pi}{8} - 4 \cos \frac{7\pi}{8}$$

Hermite Interpolation:

So far we used to see,

x	$f(x)$	$f'(x)$
3	15	40
5	27	20
9	35	37

So the degree used to be 2.

For Hermite interpolation the first derivative will also be given.

So we have additional 3 conditions, so now we will have degree 5. ($3+3=6$, so $n-1=5$)

P_{2n+1}
↓
previous
degree

$$\begin{array}{ccc} n & f(n) & f'(n) \\ \hline \end{array}$$

$$n_0 \longrightarrow f(n_0) \longrightarrow f'(n_0)$$

$$n_1 \longrightarrow f(n_1) \longrightarrow f'(n_1)$$

$$n_2 \longrightarrow f(n_2) \longrightarrow f'(n_2)$$

$$n_3 \longrightarrow f(n_3) \longrightarrow f'(n_3)$$

$\therefore$ here degree = 7.

advantage with limited knowledge we can calculate ~~higher~~ better $P(n)$.

$$\text{let, } f(n) = \sin(n) \quad f'(n) = \cos(n)$$

Now,

	n	$f(n)$	$f'(n)$
n_0	0	0	1
n_1	$\pi/2$	1	0

$$P_3(n) = h_0(n)f(n) + h_1(n)f(n_1) + \hat{h}_0(n)f'(n) + \hat{h}_1(n)f'(n_1)$$

~~$P_3(n)$~~

$$h_k(n) = \left(1 - 2(n - n_k) l'_k(n_k) \right) (l_k(n))^2$$

$$\hat{h}_k(n) = (n - n_k) (l_k(n))^2$$

$$P_3(n) = 0 + h_1(n) f(n_1) + \hat{h}_1(n_1) f'(n_1) + 0$$

~~$h_1(n) =$~~

$$l_1(n) = \frac{n - n_0}{n_1 - n_0} = \frac{n - 0}{\pi/2 - 0} = \frac{2n}{\pi}$$

$$l'_1(n) = \frac{2}{\pi}$$

$$\therefore h_1(n) = \left\{ 1 - 2(n - \pi/2) \times \frac{2}{\pi} \right\} \left(\frac{2n}{\pi} \right)^2$$

$$\hat{h}_1(n) = (n - n_0) (l_1(n))^2$$

$$= (n - 0) \left(\frac{n - n_1}{n_1 - n_1} \right)^2$$

$$= n \left(\frac{n - \frac{\pi}{2}}{-\pi/2} \right)^2 = n \left(\frac{\frac{2n}{2} - \frac{\pi}{2}}{-\frac{\pi}{2}} \right)^2 \times -\frac{2}{\pi}$$

$$= n \left(- \left(\frac{2n-n}{n} \right) \right)^2$$

$$= n \left(1 - \frac{2}{n} n \right)^2$$

$$P_3(n) = \left\{ 1 - \frac{4}{n} \left(n - \frac{n}{2} \right) \right\} \left(\frac{2}{n} n \right)^2 \times 1$$

$$+ n \left(1 - \frac{2}{n} n \right)^2 \times 1$$

Example:

	n	$f(n)$	$f'(n)$
n_0	-1	1	2
n_1	0	0	2
n_2	1	1	0

degree = 5

$$P_5(n) = h_0(n) f(n_0) + h_1(n) f(n_1) + h_2(n) f(n_2)$$

$$+ \hat{h}_0(n) f'(n_0) + \hat{h}_1(n) f'(n_1) + \hat{h}_2(n) f'(n_2)$$

$$= h_0(n) f(n_0) + h_2(n) f(n_2)$$

$$+ \hat{h}_0(n) f'(n_0) + \hat{h}_1(n) f'(n_1)$$

$$l(n) = \frac{n - n_1}{n_0 - n_1} \times \frac{n - n_2}{n_0 - n_2}$$

$$h_k(n) = \left\{ 1 - 2(n - n_k) l'_k(n_k) \right\} \left(l_k(n) \right)^2$$

$$\hat{h}_k(n) = (n - n_k) \left(l_k(n) \right)^2$$

$$h(n) = \left\{ 1 - 2(n - n_0) l'_0(n_0) \right\} \left(l(n) \right)^2$$

$$l(n) = \frac{n - 0}{0 - 0} \times \frac{n - 1}{-1 - 1} = -n \times \frac{n-1}{-2} = \frac{n(n-1)}{2}$$

$$l'(n) = 2 \times \frac{n}{2} - \frac{1}{2} \Rightarrow l'(n_0) = -1 - \frac{1}{2}$$

~~$$h_k(n) = \left\{ 1 - 2(n - (-1))(-1) \right\} \left(\frac{n^2}{2} \right)^2$$~~

$$h(n) = \left\{ 1 - 2(n - (-1)) \left(-1 - \frac{1}{2} \right) \right\} \left(\frac{n^2 - n}{2} \right)^2$$

$$= \left\{ 1 - 2(n+1) \times \left(-\frac{3}{2} \right) \right\} \left(\frac{n^2 - n}{2} \right)^2$$

$$= (1 + 3n + 3) \left(\frac{n^2 - n}{2} \right)^2$$

$$h_2(n) = \left\{ 1 - 2(n - n_2) l_2'(n_2) \right\} \left(l_2(n) \right)^2$$

$$l_2(n) = \frac{n - n_0}{n_2 - n_0} \times \frac{n - n_1}{n_2 - n_1}$$

$$= \frac{n+1}{1+1} \times \frac{n-0}{1-0} = \frac{n+1}{2} \times n = \frac{n^2+n}{2}$$

$$l_2'(n) = n + \frac{1}{2} \Rightarrow l_2'(n_2) = 1 + \frac{1}{2}$$

$$h_2(n) = \left\{ 1 - 2(n-1) \left(1 + \frac{1}{2} \right) \right\} \left(\frac{n^2+n}{2} \right)^2$$

$$= (1 - 3n + 3) \left(\frac{n^2+n}{2} \right)^2$$

$$\hat{h}_2(n) = (n - n_0) \left(l_2(n) \right)^2$$

$$= (n+1) \left\{ \frac{n(n-1)}{2} \right\}^2$$

$$\hat{h}_1 = (n - n_1) (d_1(n))^2$$

$$= n \left(\frac{n - n_0}{n_1 - n_0} \times \frac{n - n_2}{n_1 - n_2} \right)^2$$

$$= n \left(\frac{n+1}{1} \times \frac{n-1}{-1} \right)^2$$

$$= n \left(\frac{n^2 - 1}{-1} \right)^2$$

$$= n (1 - n^2)^2$$

$$\cancel{P_5(n) = \frac{1}{2} (n - n_1)}$$

$$\cancel{P_5(n) = \frac{(4+3n)(n^2-n)}{2} \times 1 + \frac{(4-3n)(n^2+n)}{2}}$$